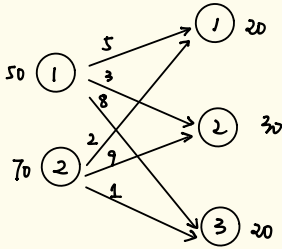


Supplement on the Transportation problem

• Unbalanced transportation problem $\sum_{i=1}^m a_i \neq \sum_{j=1}^n b_j$

- If the total supply is greater than the total demand, we add a dummy destination with 0 shipping costs. The goods to be shipped to the dummy destination will not be shipped.

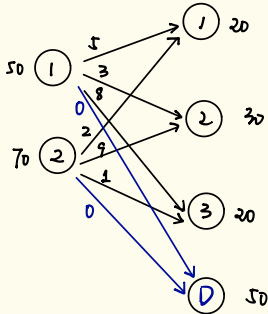
eg



$$\text{total supply} = 50 + 70 = 120$$

$$\text{total demand} = 20 + 30 + 20 = 70$$

We add a dummy destination D with demand 50 and zero shipping costs.



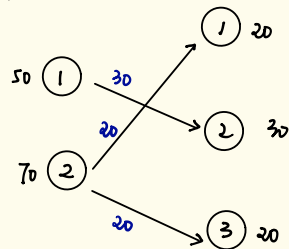
Then the transportation tableau is

	20	30	20	50
50	5	3	8	0
70	2	9	1	0

After solving the problem, we get

	20		30		20		50	
50	5	3	3		8	7	0	
70	2		9	6	1		0	
	20				20		30	

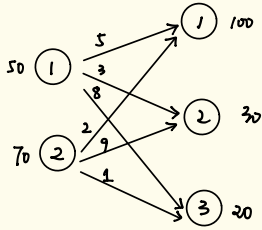
The optimal plan is:



Remark: The dummy destination brings in slack variables.

- If the total supply is less than the total demand, we add a dummy source with 0 shipping costs. The goods to be shipped from the dummy source will not be shipped.

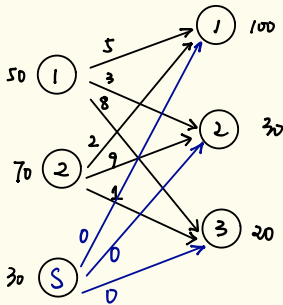
eg.:



$$\text{total supply} = 50 + 70 = 120$$

$$\text{total demand} = 100 + 30 + 20 = 150$$

We add a dummy source S with supply 30 and zero shipping costs.



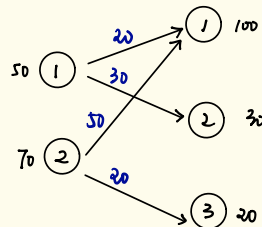
The transportation tableau is:

	100	30	20
50	5	3	8
70	2	9	1
30	0	0	0

After solving the problem, we have

	100	30	20
50	5	3	8
70	2	9	1
30	0	0	0

The optimal shipping plan is:



Remark: The dummy source brings n slack variables.

- Unimodularity

Def: Let A be an $m \times n$ matrix with integer entries. A is unimodular if $\det(B) \in \{0, 1, -1\}$ for every $m \times m$ submatrix B of A .

e.g.: $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ is unimodular.

Recall the (reduced) equality constraints $Ax = b$ of the transportation problem.

$$\begin{array}{ccccccc}
 x_{11} + x_{12} + \dots + x_{1n} & & & & & & = a_1 \\
 & x_{21} + x_{22} + \dots + x_{2n} & & & & & = a_2 \\
 & & \ddots & & & & \vdots \\
 & & & x_{m1} + x_{m2} + \dots + x_{mn} & & & = a_m \\
 x_{11} & & + x_{21} & & + x_{m1} & & = b_1 \\
 & x_{12} & & + x_{22} & & + x_{m2} & = b_2 \\
 & & \ddots & & & \ddots & \vdots \\
 & & & x_{1n} & & + x_{2n} & & + x_{mn} = b_n
 \end{array}$$

Prop. The (reduced) coefficient matrix A of the transportation problem is unimodular.

Thm: Let A be an $m \times n$ matrix with $\text{rank}(A) = m \leq n$, and suppose that the entries in A and b are all integers.

If A is unimodular, then every basic solution of $Ax = b$ is integral.

Corollary. If the data $(a_i, i=1, \dots, m; b_j, j=1, \dots, n)$ for a transportation problem are integral, then any of its basic solution is integral.